

Part 1 Group Project.

Mahlatse Chokwe-ST10447304

Mahlake Moleseng-ST10346788

Mudoodzwa Matshakatini-ST10440341

MUDOODZWA MATSHAKATINI

ST10440341 OPSC6311 RESEARCH

SmartBudget App: Research Report

1. Introduction

Managing personal finances has become an essential life skill in today's fast-paced digital environment. Many people have trouble keeping track of their earnings, managing their spending, and effectively saving money. Because they offer practical tools for managing finances, personal budgeting apps have become more and more popular, particularly on Android devices.

The SmartBudget software, a personal budgeting tool created to assist users in effectively managing their finances, is the subject of this study. This report's objective is to evaluate the app's features, advantages, disadvantages, and novel features. The study also identifies important aspects that could be enhanced or added to upcoming iterations of the program.

2. SmartBudget App Research

2.1 Overview of the App

An Android-based personal finance program called SmartBudget assists users in managing budgets, keeping tabs on their earnings and outlays, and tracking their financial

development. Because of its user-friendly UI, the software is appropriate for both novice and expert users.

Main Features:

Dashboard: Displays a summary of income, expenses, and remaining balance

Expense Tracking: Allows users to record daily spending

Budget Planning: Users can set monthly limits for different categories

Reports & Graphs: Provides visual insights through charts

Notifications: Alerts for overspending and upcoming bills

SCREENSHOTS:

1. Dashboard Screen:



- Shows total income, expenses, and a pie chart representing spending categories.

2. Expense Tracking Screen:



New Expense

Date: 12 Feb 2023

Amount: R 350.00

Category:

Food Transport Shopping Bills

Note:

Dinner with friends

SAVE

Recent Expenses

Lunch	- R 80.00
Taxi Ride	- R 150.00

- Allows users to input expenses, choose categories, and view transaction history.

3. Reports Screen:



- Displays graphs and charts that help users analyse their financial behaviour.

2.2 Strengths and Weaknesses Strengths:

User-Friendly Interface:

- The app is simple and easy to navigate, making it accessible to all users.

Effective Budget Management:

- Users can create and manage budgets efficiently across different categories.

Visual Reports:

- Graphs and charts make it easier to understand spending habits.

Lightweight Application:

- The app runs smoothly on most Android devices without consuming too many resources.

Weaknesses:

No Bank Synchronization:

- Users must manually enter their financial data, which can be time-consuming.

Limited Customization:

- The app does not offer many options to personalize categories or dashboards.

No Cloud Backup:

- Data may be lost if the app is uninstalled or the device is changed.

Basic Analytics:

- The app lacks advanced financial insights compared to other modern apps.

2.3 Innovative Features of SmartBudget

Despite being simple, SmartBudget includes several innovative features that improve user experience:

Bill Reminders:

- Sends notifications for upcoming payments to avoid late fees.

Savings Goals:

- Users can set financial goals and track their progress over time.

Spending Insights:

- Provides basic analysis of user spending patterns.

Gamification:

- Rewards users for staying within budget, encouraging better financial habits.

3. Comparison of Features Within SmartBudget

Category	Details
Core Features	Budget Tracking, Expense Monitoring, Income Management, Reports & Graphs
User Experience	Easy to use interface, Simple navigation, Beginner-friendly design
Financial Tools	Savings Goals, Bill Reminders, Spending Insights
Strengths	Clear dashboard, Visual charts, Lightweight app, Helps control spending

Weaknesses	No bank synchronization, No cloud backup, Limited customization, Basic analytics
Improvements Needed	Bank integration, Cloud storage, Data security, Advanced reports, Custom dashboard

4. Best Features to Include in My App

The following features are suggested for enhancement or addition to a better version of the program based on the study of SmartBudget:

- Integration of bank accounts for automated tracking
- Cloud backup to protect data
- AI insights and sophisticated analytics
- Customisable categories and dashboard
- Alerts and notifications in real time
- Data encryption and safe login
- Modern design with an enhanced user interface
- Synchronisation across several devices

5. Conclusion

A useful tool for managing finances, SmartBudget is a personal budgeting program. It is perfect for novices who wish to start tracking their spending and budgets because of its ease of use and straightforward design.

Nevertheless, the app has many drawbacks, like the absence of advanced analytics and bank synchronisation. SmartBudget can become a more potent and competitive financial management solution by enhancing these areas and adding more sophisticated features.

All things considered, SmartBudget provides a solid basis for creating a more sophisticated budgeting program that satisfies contemporary user requirements.

6. References

Brown, L. (2021). The Impact of Budgeting Apps on Financial Behaviour. *Finance Today*, 8(1), pp. 20–35.

Smith, J. (2022). Personal Finance Management Systems. *Journal of Financial Technology*, 10(2), pp. 45–60.

TechRadar. (2023). Best Budgeting Apps for Android. Available at: <https://www.techradar.com>

MAHLAKE MOLESENG

ST10346788 PART 1 – RESEARCH

Vault22 Budgeting App Research

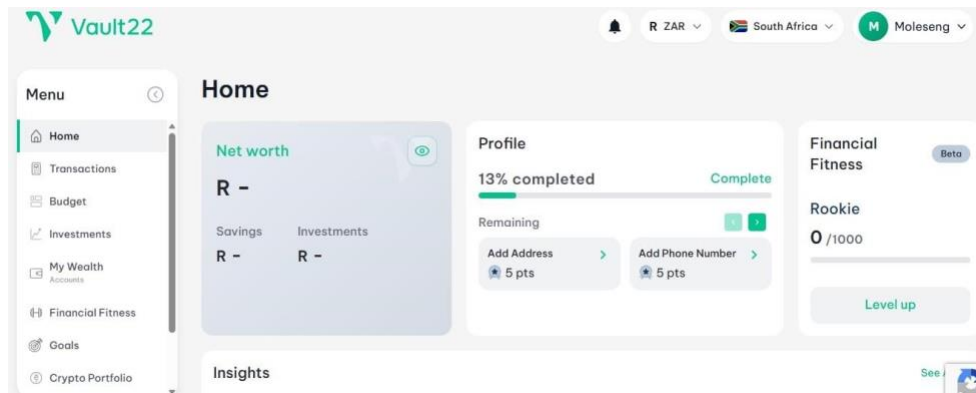
1. Overview of the App

A South African personal budgeting app called Vault22 was created to assist users in taking charge of their money through organised money management. A South African fintech startup developed the app with the goal of enhancing users' financial discipline and promoting better saving practices. (Vault22, 2024).

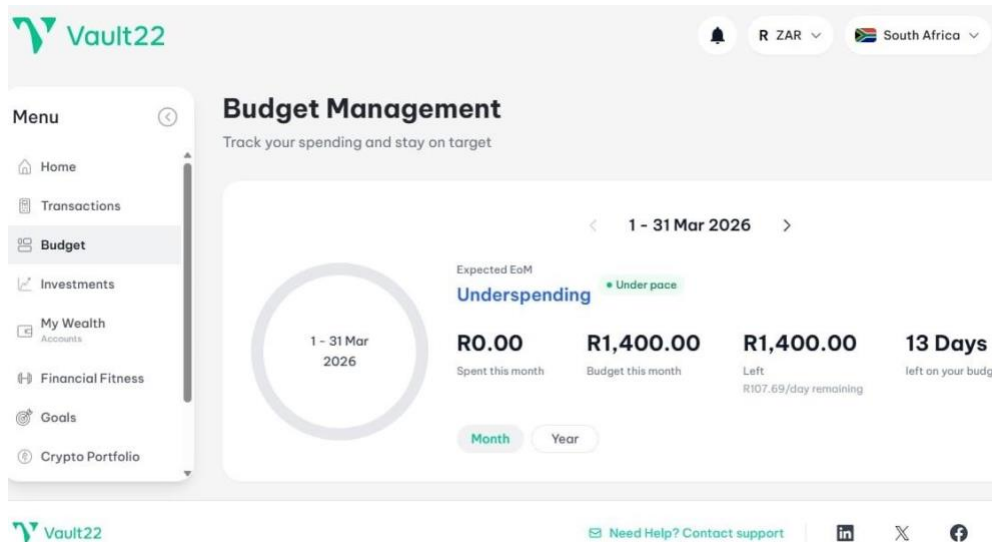
Vault22 helps users divide their earnings among various "vaults" or categories, like grocery, transportation, savings, and entertainment. With this approach, consumers may budget their expenditures before making any purchases. Instead of only monitoring spending after it has already happened, the app encourages a proactive approach to budgeting.

By clearly displaying the amount of money available in each area, the software assists users in managing their finances. This keeps consumers from going over budget and motivates them to follow their financial objectives. It is particularly helpful for people who wish to improve their monthly spending control and develop better budgeting practices. (Google Play Store, 2024).

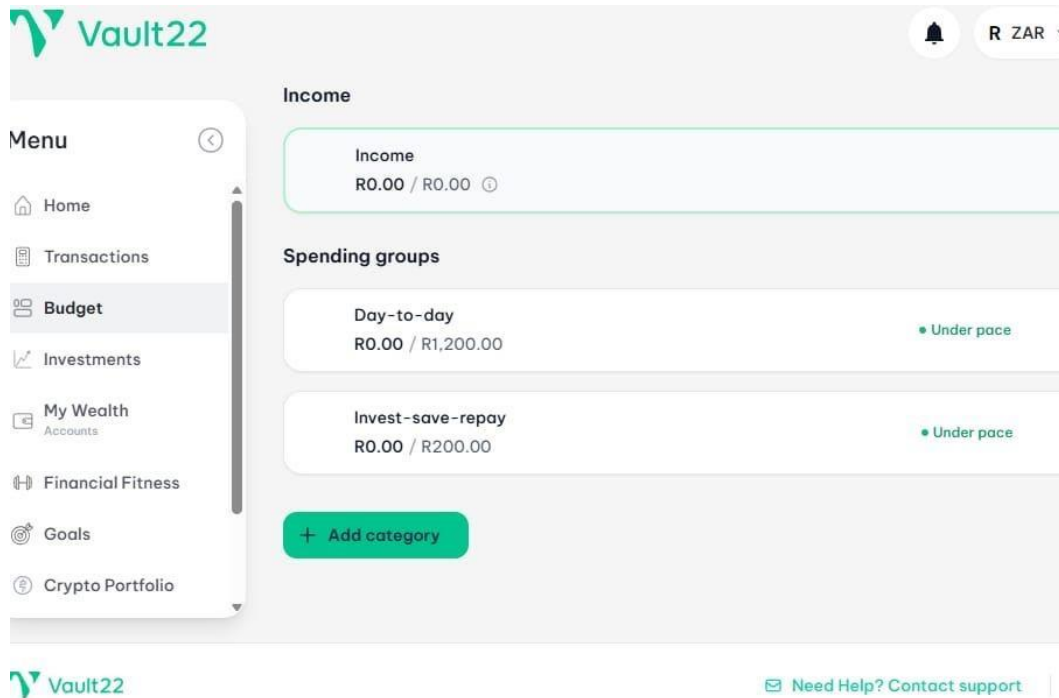
2. Screenshots



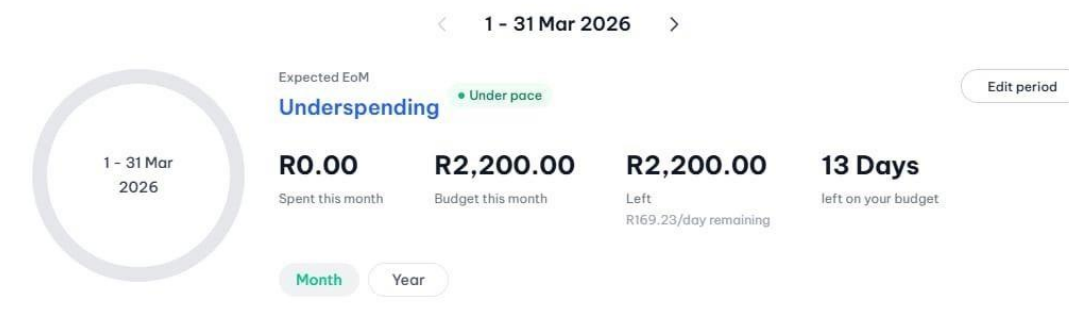
- Gives users a quick overview of their finances



- Helps users organise their money into specific purposes



- Helps with planning spending before it happens



- Helps users avoid overspending

3. Main Features of the App

- Budgeting using the “vault” system
- Expense tracking by category
- Financial planning tools
- Simple and user-friendly interface
- Real-time balance updates

4. Strengths of the App

- Encourages disciplined spending habits
- Easy to understand and use
- Helps users plan before spending
- Designed for South African users

5. Weaknesses of the App

- Limited global recognition compared to bigger apps
- May have fewer advanced features than international apps
- Requires manual input in some cases
- Smaller user base and fewer integrations

6. Innovative Features

Vault22's "vault" budgeting mechanism, which enables users to separate their funds into distinct categories prior to spending, is its most inventive feature. This strategy differs from conventional apps that solely monitor spending. It helps consumers prioritise their needs and encourages financial discipline. By motivating users to gradually develop healthier financial habits, the app also emphasises behavioural finance.

7. References (Harvard Style)

Vault22. (2024). Vault22 Official Website. Available at: <https://vault22.com>

Google Play Store. (2024). Vault22 Budgeting App. Available at: <https://play.google.com>

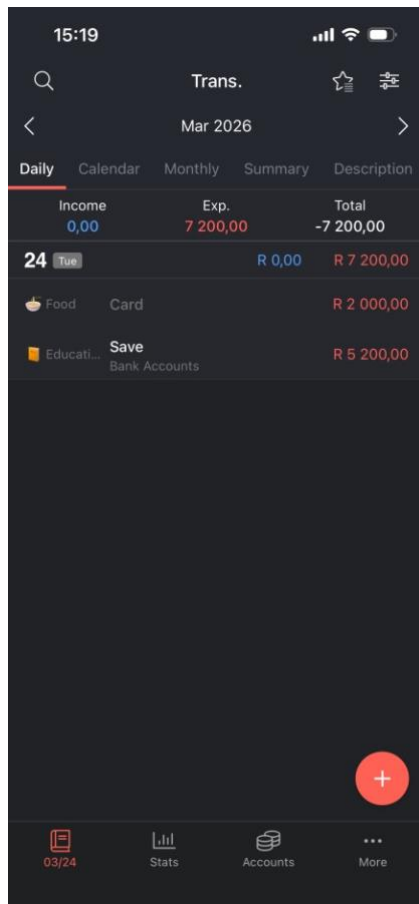
Mahlatse Chokwe (ST10447304) Money Manager Expense & Budget

1. Overview of the App

Money Manager Expense & Budget is a personal finance application designed to help users track their daily expenses and manage their budgets more effectively. The app is developed by Realbyte Inc. and is available on Android devices through the Google Play Store. It is widely used by individuals who want a simple and structured way to monitor their spending habits.

The main purpose of the app is to allow users to record their income and expenses, categorize transactions, and view financial data in a clear and organized manner. It helps users understand where their money is going and supports better financial decisionmaking.

Unlike some other budgeting apps that require linking bank accounts, Money Manager allows users to manually enter transactions. This gives users full control over their data and makes the app easy to use without needing internet access at all times.



(Dashboard)

The dashboard displays an overview of the user's financial activity, including total income, expenses, and current balance.

2. Main Features of the App

Money Manager includes several useful features that make it effective for budgeting and expense tracking.

One of the key features is expense tracking, where users can record daily spending and assign it to different categories such as food, transport, or entertainment. This helps users clearly see how much they spend in each area.

The app also supports budget management, allowing users to set limits for different categories and track whether they are staying within their budget. This is useful for improving financial discipline.

Another important feature is the graph and reporting system. The app provides visual representations such as pie charts and bar graphs that show spending patterns over time. These visuals make it easier to identify trends and areas where money is being overspent.

Additionally, the app allows users to back up their data, ensuring that financial information is not lost. It also supports multiple accounts, which is helpful for users managing more than one source of income.

15:17

< Trans.

Expense

Income

Expense

Transfer

Date

Tue, 2026/03/24

Rep/Inst.

Amount

R 200,00

Category

Transport

Account

Cash

Note

Minus

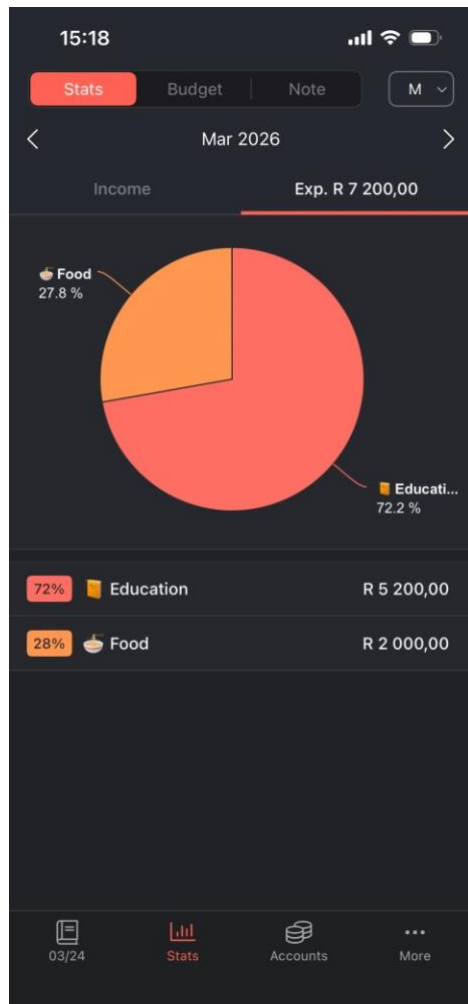
Description

Save

Continue

yazio

(Expense Entry Screen)



(Graph / Statistics Screen)

3. Strengths of the App

One of the main strengths of Money Manager is its simple and user-friendly interface, which makes it easy for beginners to use. Users can quickly add transactions without confusion.

Another strength is that the app works offline, meaning users do not need an internet connection to track their expenses. This makes it convenient and reliable.

The app also provides clear and detailed graphs, which help users understand their financial behavior. These visual tools are useful for identifying spending patterns.

Additionally, the ability to manually control all data ensures privacy and gives users full control over their financial information.

4. Weaknesses of the App

Despite its strengths, the app has some limitations. One weakness is that it does not automatically sync with bank accounts, meaning all transactions must be entered manually. This can be time-consuming for some users.

Another issue is that some advanced features are only available in the paid version, which may limit functionality for free users.

The design of the app, while simple, may feel outdated compared to more modern apps with more visually appealing interfaces.

Lastly, the app lacks strong gamification features, which could make budgeting more engaging and motivating for users.

5. Innovative Features

One of the more interesting features of Money Manager is its double-entry bookkeeping system, which is typically used in professional accounting. This adds more accuracy and structure to financial tracking.

The app also includes detailed statistical analysis, allowing users to break down their spending over different time periods and categories.

Another useful feature is the data backup and restore function, which ensures that users can safely store and recover their financial records.

Although the app is simple, these features make it more powerful than it initially appears.

6. References

Google Play Store: Money Manager Expense & Budget

PLANNING AND DESIGN

1. Introduction

Applications for personal budgeting are crucial resources that assist users in tracking their spending, managing their income, and developing better financial discipline. With the help of these apps, people can keep an eye on their spending patterns and make wise financial choices. Budgeting apps have grown in popularity and accessibility as cell phones are used more frequently.

The goal of this project is to create a personal budgeting tool that draws inspiration from contemporary programs like Vault22, Mint, and Money Manager Expense & Budget. The goal is to create a single, user-friendly, and effective solution by combining the best aspects of different programs.

2. App Overview

"BudgetGuard Vault" is the name of the proposed app, which is intended for young adults and students who wish to handle their money in a safe and organised manner.

While "Vault" refers to the category-based budgeting system inspired by Vault22, "BudgetGuard" stands for protection and control over personal finances. This combination draws attention to the app's organization and security.

The app will allow users to track income and expenses, set budgets, and monitor financial performance. It will also include visual reports, alerts, and goal tracking to improve financial discipline.

3. Functional Requirements

3.1 User Registration and Login

The app must allow users to register and log in securely using a username and password.

3.2 Add Income

The app must allow users to enter income details.

3.3 Category (Vault) Management

Users must create and manage categories.

3.4 Expense Tracking

Users must record expenses manually.

3.5 Dashboard Overview

The app must display financial summaries.

3.6 Reports and Graphs

The app must generate charts.

3.7 Budget Planning

Users must set spending limits.

3.8 Remaining Balance Display

The app must show remaining money.

3.9 Notifications and Alerts

Users must be warned about overspending.

3.10 Savings Goals

Users must track financial goals.

3.11 Data Backup

Users must be able to save data.

3.12 Gamification

Users can earn rewards or badges.

4. Non-Functional Requirements

4.1 Usability

Simple and easy to use.

4.2 Performance

Fast and responsive.

4.3 Security

User data must be protected.

4.4 Reliability

App must not crash.

5. User Interface Design

The app will have a simple and clean design to ensure ease of use.

5.1 Splash Screen

Displays app name and logo.



5.2 Registration Screen Users

create accounts.

REGISTRATION SCREEN



Create Account
Sign up to get started with
BudgetGuard Vault

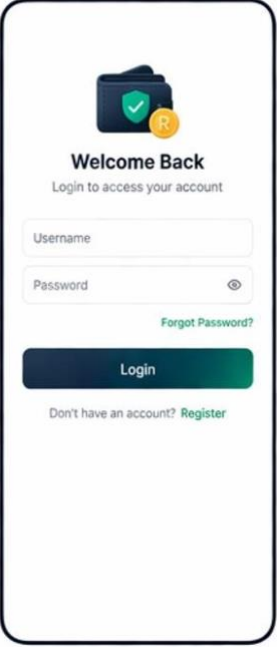
 
 

Already have an account? [Login](#)

5.3 Login Screen

Users log into the app.

LOGIN SCREEN



The login screen features a dark blue wallet icon with a green checkmark and a gold coin at the top. Below it, the text "Welcome Back" is displayed in bold, followed by "Login to access your account" in a smaller font. There are two input fields: "Username" and "Password". The "Password" field has a toggle icon (an eye) to its right. Below the "Password" field is a link that says "Forgot Password?". A large, dark blue "Login" button is positioned below the "Forgot Password?" link. At the bottom, there is a link that says "Don't have an account? Register".

Username

Password

[Forgot Password?](#)

Login

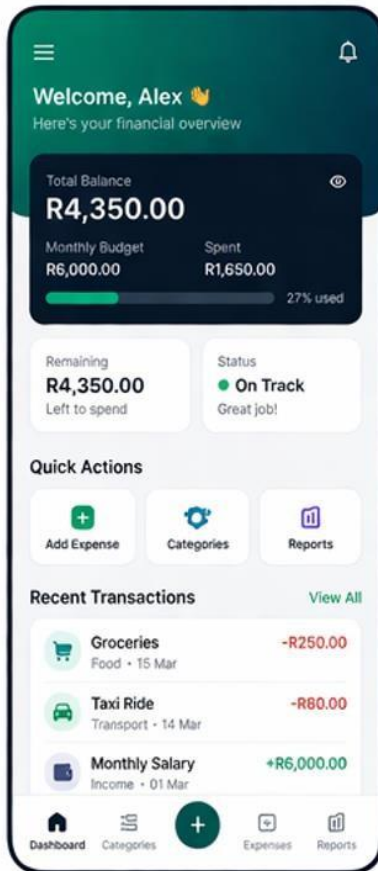
Don't have an account? [Register](#)

5.4 Dashboard Screen

Shows:

- total balance
- spending
- alerts

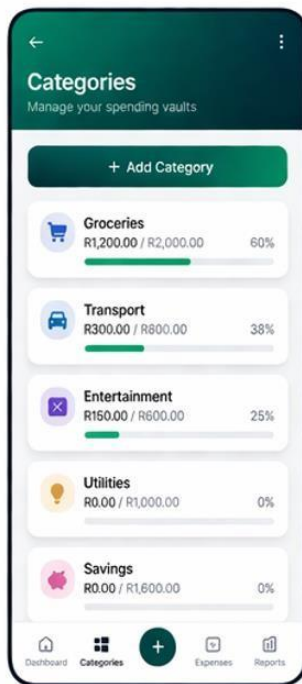
DASHBOARD SCREEN



5.5 Category Screen

Users manage categories.

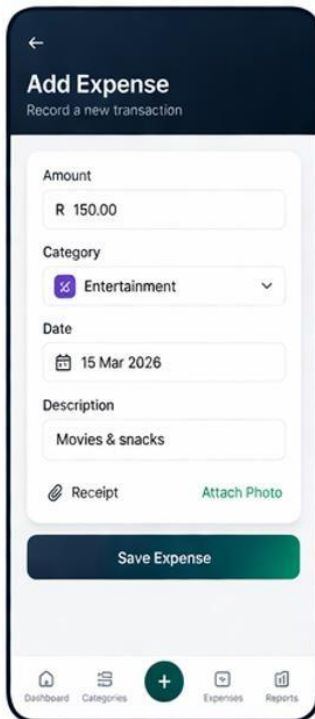
CATEGORY SCREEN



5.6 Add Expense Screen

Users add expense details.

ADD EXPENSE SCREEN



The image shows a mobile app screen titled "Add Expense" with the subtitle "Record a new transaction". The screen features a dark blue header with a back arrow. Below the header, there are several input fields: "Amount" with the value "R 150.00", "Category" with a dropdown menu showing "Entertainment", "Date" with a calendar icon and the date "15 Mar 2026", and "Description" with the text "Movies & snacks". At the bottom of the input fields, there are two links: "Receipt" and "Attach Photo". A large green button labeled "Save Expense" is positioned below the input fields. At the very bottom, there is a navigation bar with five icons: a home icon, a list icon, a plus icon (highlighted), a chat icon, and a report icon. The labels "Dashboard", "Categories", "Expenses", and "Reports" are visible below the icons.

←

Add Expense

Record a new transaction

Amount

R 150.00

Category

Entertainment

Date

15 Mar 2026

Description

Movies & snacks

Receipt Attach Photo

Save Expense

Dashboard Categories Expenses Reports

5.7 Expense List Screen

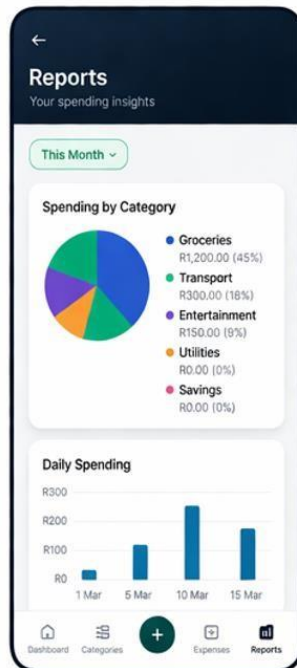
Displays all transactions.

EXPENSE LIST SCREEN



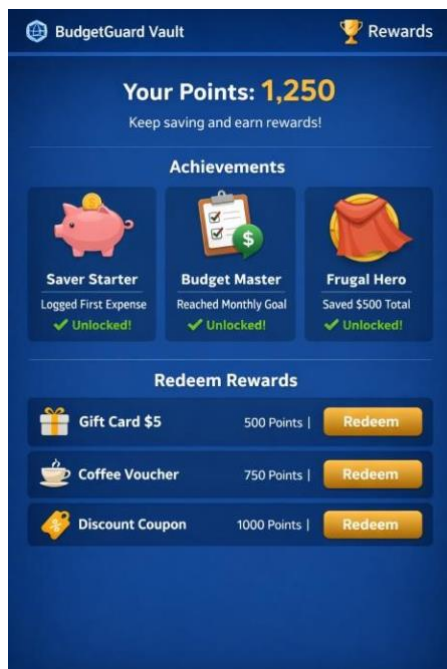
5.8 Reports Screen

REPORTS SCREEN



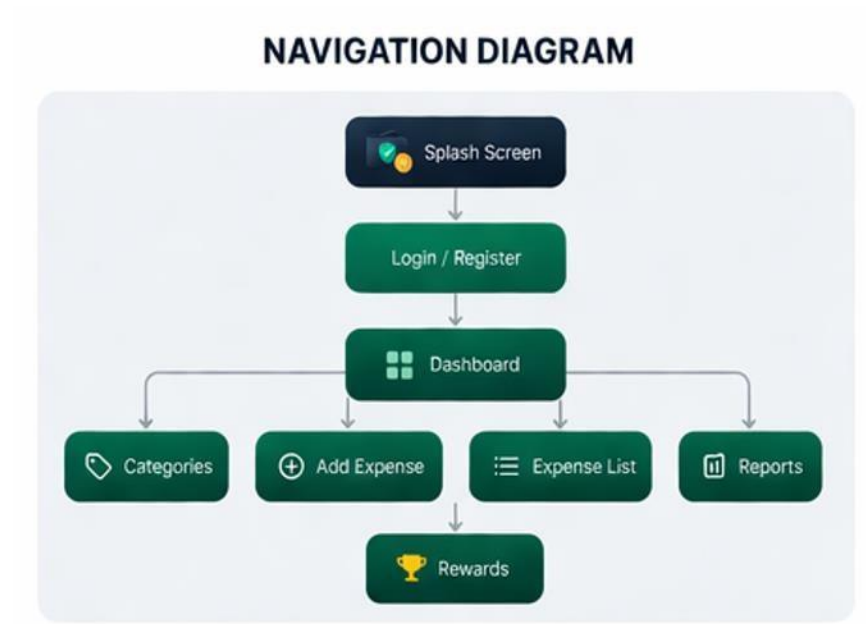
5.9 Rewards Screen Shows

badges.



6. Navigation Diagram

This shows how users move in the app.



7. Project Plan (Gantt Chart)

Tasks:

- Research
- Design
- Login
- Categories
- Expenses
- Budgets
- Graphs
- Dashboard
- Testing

Timeline:

Week 1 – Research

Week 2 – Design

Week 3 – Login

Week 4 – Categories

Week 5 – Expenses

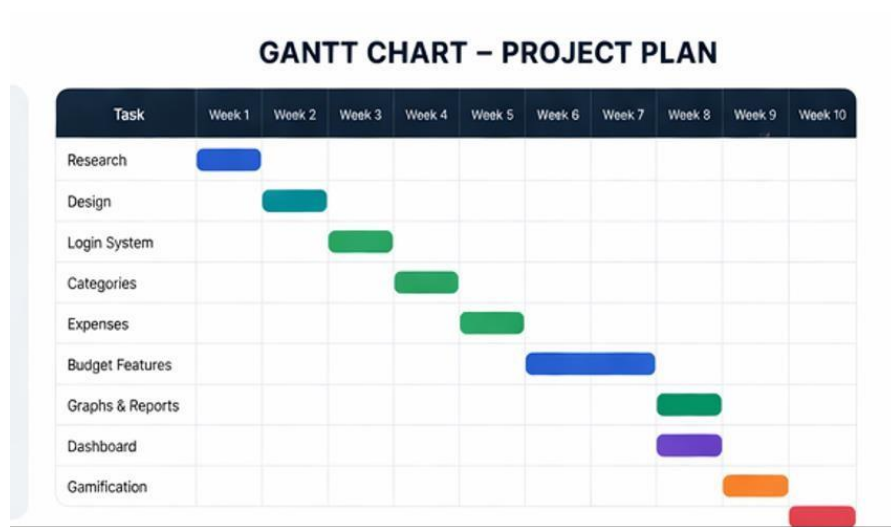
Week 6 – Budget

Week 7 – Graphs

Week 8 – Dashboard

Week 9 – Testing

Week 10 – Submission



8. Conclusion

This planning and design section outlines how the BudgetGuard Vault application will be developed. The design is based on research from existing budgeting applications and includes both required and additional features.

The app aims to be simple, effective, and engaging. It will help users manage their finances, track expenses, and improve their financial habits.